When Multigranularity Meets Spatial–Spectral Attention: A Hybrid Transformer for Hyperspectral Image Classification

Er Ouyang, Bin Li, Wenjing Hu, Guoyun Zhang, *Member, IEEE*, Lin Zhao, *Member, IEEE*, and Jianhui Wu

***Abstract*— The transformer framework has shown great potential in the field of hyperspectral image (HSI) classification due to its superior global modeling capabilities compared to convolutional neural networks (CNNs). To utilize the transformer to model spatial–spectral information, a hybrid transformer that integrates multigranularity tokens and spatial–spectral attention (SSA) is proposed. Specifically, a token generator is designed to embed the multigranularity semantic tokens, which contributes richer image features to the model by exploiting CNN's local representation capability. Moreover, a transformer encoder with an SSA mechanism is proposed to capture the global dependencies between different tokens, enabling the model to focus on more differentiated channels and spatial locations to improve the classification accuracy. Ultimately, adaptive weighted fusion is applied to different granularity transformer branches to boost HybridFormer's classification performance. Experiments were conducted on four new challenging datasets, and the results indicate that HybridFormer achieves state-of-the-art results in terms of classification performance. The code of this work will be available at https://github.com/zhaolin6/HybridFormer for the sake of reproducibility.**

***Index Terms*— Attention mechanism, convolutional neural networks (CNNs), hyperspectral image (HSI) classification, multigranularity, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain more continuous spectral bands than conventional RGB images and are rich in spatial–spectral detail information. Benefiting from this advantage, HSIs have been successfully used in a wide range of applications in many fields, including environmental monitoring [1], plant disease monitoring [2], fine agriculture [3], and earthquake prediction [4]. HSI classification has always been one of the most crucial aspects of HSI data processing, and it aims to identify different areas of the same land cover in a multidimensional feature space. However, as HSI acquisition is often disturbed by instrumental effects [5] and noise [6], accurately identifying land-cover classes has attracted considerable attention.

Recently, deep convolutional neural networks (CNNs) have become the mainstream method for HSI classification, and their performance exceeds that of most traditional machine learning-based methods due to their ability to learn deep abstract features [7]. Hu et al. [8] designed a 1-D-CNN network architecture, which directly extracts depth features in the spectral domain to achieve HSI classification, and Zhao and Du [9] proposed the spatial-based 2-D-CNN for extracting the spatial and spectral features of the main components. Zhao et al. [10] proposed a spectral attention-based 2-D-CNN network to capture the spectral information of HSI. To address the specific information that arises from the coexistence of spatial–spectral information in the HSIs of 3-D data, Hamida et al. [11] proposed a 3-D-CNN network that enables the joint processing of spatial and spectral data. Roy et al. [12] introduced a hybrid 3-D and 2-D spectral CNN (HybridSN) model for HSI classification that combines complementary spatial–spectral and spectral information, and it achieves extremely high classification accuracy. Zheng et al. [13] proposed a mixed 3-D–2-D convolution and covariance pooling embedding into a conventional CNN architecture to extract more discriminative spectral and spatial features.

In addition, numerous other deep learning-based methods have been successfully applied to HSI classification [14], [15]. These methods include generative adversarial networks (GANs) [16], [17], graph convolutional networks (GCNs) [18], [19], recurrent neural networks (RNNs) [20], [21], stacked autoencoders (SAEs) [22], [23], deep belief networks (DBNs) [24], [25], and capsule networks [26], [27]. Among the above network, CNNs have remained an integral and dominant part of HSI classification. CNN architectures capture local information by using local receptive fields, shared parameters, and spatial subsampling [28], which have better generalization capabilities.

Although CNN-based methods have achieved excellent results on HSI classification tasks, some defects still exist. First, the receptive field of CNN-based methods is restricted to a square region of the convolutional kernel size due to the

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kernel's limitations, which usually renders the network inept in terms of modeling long-distance dependencies [29]. Second, traditional CNN-based methods cannot effectively extract features from HSI when they contain complex land-cover structures with different scales or shapes [30].

More recently, a deep learning model based on the self-attention mechanism (MHSA) [31] called transformer [32] has been widely used for natural language processing (NLP). Devlin et al. [33] proposed a model called bidirectional encoder representations from transformers (BERT), which treats the input word as a continuous sequence and uses the MHSA to capture global contextual information in a large range. Dosovitskiy et al. [34] designed a vision transformer (ViT) and successfully transferred the transformer model from NLP to computer vision to explore the direct application of transformer networks in image recognition. Unlike CNN-based models with local perception, the transformer uses the MHSA to model the input image's global dependencies. Due to the successful application of transformer in computer vision, many works have attempted to apply it to the HSI classification task. The HSI-BERT model, which was proposed by He et al. [35], is the first framework to employ BERT in HSI classification, which exploits the fact that the transformer has a global field of perception to capture global dependencies between pixels. Considering the characteristics of HSI, Yu et al. [36] proposed a multilevel spectral–spatial transformer network (MSTNet) that improves the classification accuracy by fusing multilevel features. To fully exploit the local spectral–spatial semantic information, Sun et al. [37] proposed the spectral–spatial feature tokenization transformer (SSFTT) method, which adopts a Gaussian distribution weighted tokenization module to transform the shallow spatial–spectral features into tokenized semantic features. However, these transformer methods adopt fixed-size patches at the phase of patch embedding, which is prone to misclassification of boundary categories. Moreover, these methods use the global modeling capabilities of the transformer to calculate correlations between spatial locations while ignoring the abundant spectral information available in HSIs.

As a result, we propose a hybrid transformer with multigranularity spatial–spectral attention (SSA) for HSI classification (HybridFormer). The HybridFormer model focuses on exploiting the HSI's spatial and spectral information, as well as effectively combining the transformer's global modeling capability with convolution's local information perception capability to further mine the HSI's depth semantic information. First, distinguishing from other methods that use fixed-size patches to perform patch embedding, a multigranularity token generator (MTG) is proposed to generate semantic tokens with different sizes while ensuring the independence of spatial and spectral information. Second, a hybrid transformer encoder structure is designed to learn the long-distance dependencies between semantic tokens to exploit the hybrid spatial and spectral information using the proposed SSA module. Finally, a weighted adaptive fusion method is employed to balance the information from different granularity branches and enhance the HybridFormer's discriminability.

The primary contributions of this article are listed as follows.

1) We investigate the HSI classification problem from the perspective of multigranularity semantic token embedding and spatial–spectral feature modeling. HybridFormer is proposed as a novel transformer-based backbone network, which utilizes a local–global–local strategy to efficiently model the spatial and spectral information of HSI. HybridFormer effectively mixes the transformer's global long-distance dependence with CNN's local modeling capability to fully exploit features and further enhance the classification accuracy.
2) A novel SSA is proposed to replace the self-attention mechanism in the original transformer, and it captures the HSI's spatial–spectral information by calculating the correlation between the spatial position and spectral so that HybridFormer can focus on more differentiated features and, thereby, improve the classification accuracy.
3) An MTG is proposed, which performs patch embedding through depthwise convolution so that the spatial and spectral information remains independent. The multigranularity information is input into the subsequent transformer module, which can further improve the discriminability of the HybridFormer.

The rest of this article is organized as follows. Section II describes the proposed HybridFormer framework. Section III provides the experimental results obtained with four datasets and further analyzes the proposed modules. Finally, Section IV concludes this article and discusses possible future improvements.

II. Methods

A. Proposed HybridFormer Architecture

As shown in Fig. 1, the overall structure of the HybridFormer model consists of three main components: MTG, transformer encoder, and classifier. Specifically, the input of the HybridFormer is fixed-size patch data, which is fed into the MTG. The CNN feature extractor extracts the shallow semantic information, and these features are further input into the multigranularity depthwise convolution token embedding (MDTE) module to tokenize the coarse-grained and fine-grained subpatches. Then, these tokens are sent to the hybrid transformer encoder to learn the long-distance dependencies, which primarily consists of the SSA and the feed-forward neural network (FFN). During the classification stage, the features of different granularities that were captured by the transformer are input into the linear layer, and then, an adaptive weighted fusion module is proposed to fusion these features. Ultimately, the output with the highest probability value is the classification result.

B. Multigranularity Token Generator

Most of the existing transformer-based methods use fixed-size subpatches and ignore the multigranularity information of the HSI; here, the granularity refers to the size of the subpatch. In this article, to obtain more abundant feature

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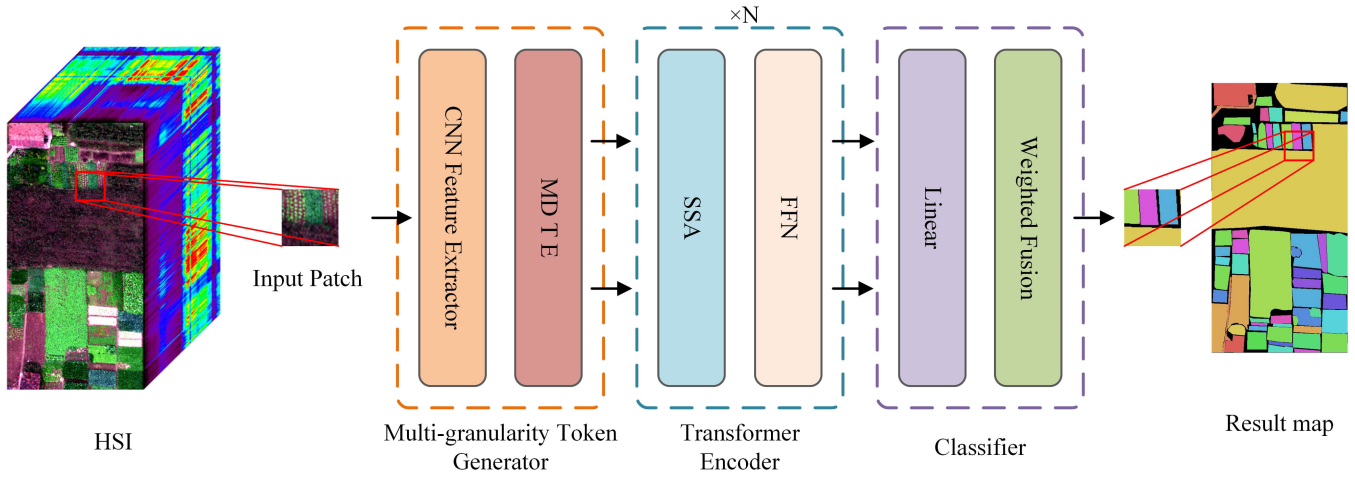


Fig. 1. Architecture of the proposed HybridFormer.

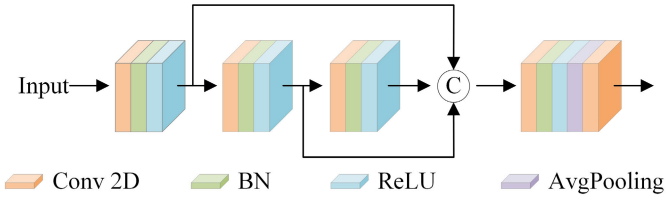


Fig. 2. Structure of the CNN feature extractor.

information, an MTG is proposed. The MTG consists of a CNN feature extractor and an MDTE module, which uses the CNN to extract abstract semantic features and embeds tokens at different granularities and with completely separated spatial–spectral information into the transformer. Therefore, the multigranularity features are used as input to the transformer in the HybridFormer model.

1) CNN Feature Extractor: To effectively extract local semantic information from HSI, we designed a simple CNN-based semantic information feature extractor. Due to the limited number of HSIs training samples, a simple four-layer convolution block is applied to extract the shallow features. As shown in Fig. 2, each convolutional layer is followed by a batch normalization (BN) and a rectified linear unit (ReLU). Given an HSI cube of size $W \times H \times C$, which is cropped into patches of fixed size $P \times P \times C$, these patches are first passed through three-layer convolution blocks with convolution kernel sizes of 1 to fully exploit the information in the spectral domain. After concatenating the different levels of features, a convolution layer with a kernel size of 3 captures the spatial semantic information. Finally, the average pooling layer is used to reduce the variance of data, and a convolutional layer is employed to fix the spectral dimensionality to a suitable number. Table I details the relevant parameters of the CNN feature extractor.

2) Multigranularity Depthwise Convolution Token Embedding: To fully capture the multigranularity features from the HSI, an MDTE module was designed to transform the vision feature into semantic tokens with different granularities, which enables the model to perceive HSI information at

TABLE I
NETWORK SUMMARY OF THE FEATURE EXTRACTOR ARCHITECTURE

Name	Layer	Kernel Size	Padding	With
1	Conv2D	(1, 1)	0	BN + ReLU
2	Conv2D	(1, 1)	0	BN + ReLU
3	Conv2D	(1, 1)	0	BN + ReLU
4	Conv2D	(3, 3)	1	BN + ReLU
	AvgPool2d	(3, 3)	1	-
	Conv2D	(1, 1)	0	-

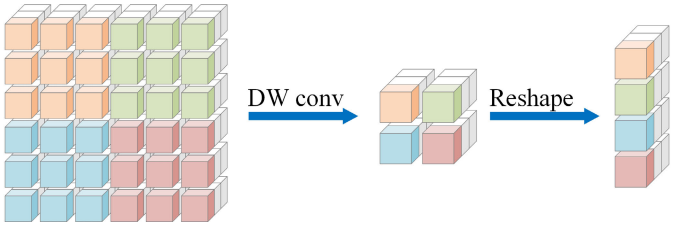


Fig. 3. Depthwise convolution token embedding module.

different granularities. In addition, the application of depthwise convolution also separates spatial and spectral features, which preserves the channel distribution information and provides a basis for the subsequent SSA mechanism. Fig. 3 illustrates the process of MDTE.

Specifically, given a feature map $I' \in \mathbb{R}^{w \times h \times c}$ that has been extracted by the CNN feature extractor, where $w \times h$ is the size of I' and c is the number of bands, depthwise convolution with different kernel sizes is used to partition the feature map into several semantic tokens with different granularities. The result T can be obtained by the following equation:

$$T_i = \mathrm{DWConv2}D_i\left(I'\right), \quad i = 1, 2, \ldots \tag{1}$$

where i represents the ith granularity branch and DWConv2D$(\cdot)$ denotes a 2-D depthwise convolution operation. It is worth noting that the granularity of each block can be adjusted by setting the depthwise convolution size. For the given feature map I', the output dimension of each

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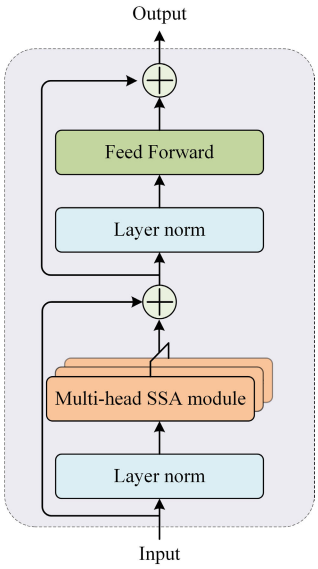


Fig. 4. Illustration of a transformer encoder.

depthwise convolution can be represented as

$$P_i = \left(\frac{w + 2 \times p_i - k_i}{s_i} + 1, \frac{h + 2 \times p_i - k_i}{s_i} + 1 \right) \quad (2)$$

where p_i, k_i, and s_i represent the padding, kernel size, and stride of the depthwise convolution in the ith granularity branch. Finally, a reshape operation is used to serialize the resulting patch matrix to obtain the token sequence, which is denoted by $T = [T_i \in \mathbb{R}^{n_i \times c}, i = 1, 2, \ldots,]$, where n_i is the number of semantic tokens corresponding to the ith granularity branch.

C. Transformer Encoder

The transformer encoder is the core of the HybridFormer model, which is used to learn global and local information from the input tokens. The transformer encoder primarily consists of the SSA module and the FFN, as shown in Fig. 4. The global dependencies between the embedded tokens are captured using the multihead SSA module, while the representation of local information is augmented by the convolutional structure of FFN. In addition, residual connections are placed in front of each layer normalization (LN) block to alleviate the problem of gradient disappearance or explosion.

1) SSA Module: To fully capture the spectral–spatial features of HSI, we proposed a novel attention module called SSA, which contains two different types of attention modules: spatial attention and spectral attention. Distinguishing from the other attention module, the SSA module captures the spatial and spectral features from local patches and channels, and it highlights discriminative features using mutual attention. Fig. 5 illustrates the overall framework of the SSA. Given a token sequence $T \in \mathbb{R}^{n \times d}$ that has been through the MTG subnetwork, where n is the number of tokens and d is the number of channels, the results of the SSA module can be represented as

$$T_{\text{SSA}} = \text{SpeAttn}\left(\text{SpaAttn}\left(T^{\top}\right)\right)^{\top} \quad (3)$$

where SpeAttn$(\cdot)$ is the spectral attention, SpaAttn$(\cdot)$ is the spatial attention, and $^{\top}$ is the transpose operation.

Specifically, spatial attention is designed to capture the relationship between tokens. $Q_{\text{spa}} \in \mathbb{R}^{n \times d}$, $K_{\text{spa}} \in \mathbb{R}^{n \times d}$, and $V_{\text{spa}} \in \mathbb{R}^{n \times d}$ are obtained by T through a linear layer, respectively. Thus, the self-attention matrix W_{spa} can be expressed as

$$W_{\text{spa}} = \text{Conv2D}\left(Q_{\text{spa}} \otimes K_{\text{spa}}^{\top}\right) \quad (4)$$

where Conv2D$(\cdot)$ is a 2-D convolutional operation, which is used to aggregate the correlation of adjacent tokens. Finally, the results of spatial attention T_{spa} can be obtained by

$$T_{\text{spa}} = \text{SpaAttn}(T) = \text{Softmax}\left(\frac{W_{\text{spa}}}{\sqrt{d_{K_{\text{spa}}}}}\right) V_{\text{spa}} \quad (5)$$

where $d_{k_{\text{spa}}}$ is the feature dimension of K_{spa} and Softmax$(\cdot)$ is the softmax function.

In addition to spatial attention, spectral attention is proposed to capture the correlation between the spectral dimensions of tokens. Specifically, a transpose operation is used to transpose T_{spa} to obtain $T_{\text{spa}}^{\top} \in \mathbb{R}^{d \times n}$. Then, $Q_{\text{spe}} \in \mathbb{R}^{d \times n}$, $K_{\text{spe}} \in \mathbb{R}^{d \times n}$, and $V_{\text{spe}} \in \mathbb{R}^{d \times n}$ are each obtained from $T_{\text{spa}}^{\top}$ through a linear layer, respectively. Thus, the spectral attention matrix W_{spe} can be obtained by the following equation:

$$W_{\text{spe}} = \text{Conv2D}\left(Q_{\text{spe}} K_{\text{spe}}^{\top}\right). \quad (6)$$

The results of spectral attention T_{spa} can be expressed by

$$T_{\text{spe}} = \text{SpeAttn}\left(T_{\text{spa}}\right) = \text{Softmax}\left(\frac{W_{\text{spe}}}{\sqrt{d_{K_{\text{spe}}}}}\right) V_{\text{spe}} \quad (7)$$

where $d_{K_{\text{spe}}}$ is the feature dimension of K_{spe} and $T_{\text{spe}} \in \mathbb{R}^{d \times n}$. Finally, a transpose operation is used to transpose the token sequences $T_{\text{spe}} \in \mathbb{R}^{d \times n}$ to $T_{\text{SSA}} \in \mathbb{R}^{n \times d}$. After completing the above steps, the model can effectively capture the global dependencies between different tokens, which further improves the performance of the model. After completing the above steps, the model can effectively capture the global dependencies between different tokens, which further improves the performance of the model.

2) Feed-Forward Neural Network: To further extract the local feature information of HSI, we proposed an FFN, which is composed of a depthwise separable convolution and two 2-D convolutions. For the output $T_{\text{SSA}} \in \mathbb{R}^{n \times d}$ of the SSA module, a reshape operation is first used to resize the dimension $n \times d$ of T_{SSA} into $w \times h \times d$, where $n = w \times h$. Then, a depthwise separable and two 2-D convolutions are used to extract the spatial–spectral information of HSI. Finally, the dimensionality of the features is restored to $n \times d$ in preparation for the next transformer encoder module.

D. Classifier

During the final classification stage, there are multiple feature matrices from different granularity branches that are obtained by the transformer. These feature matrices are fed separately into a classifier, which is composed of a fully connected layer, to calculate the probability that the extracted

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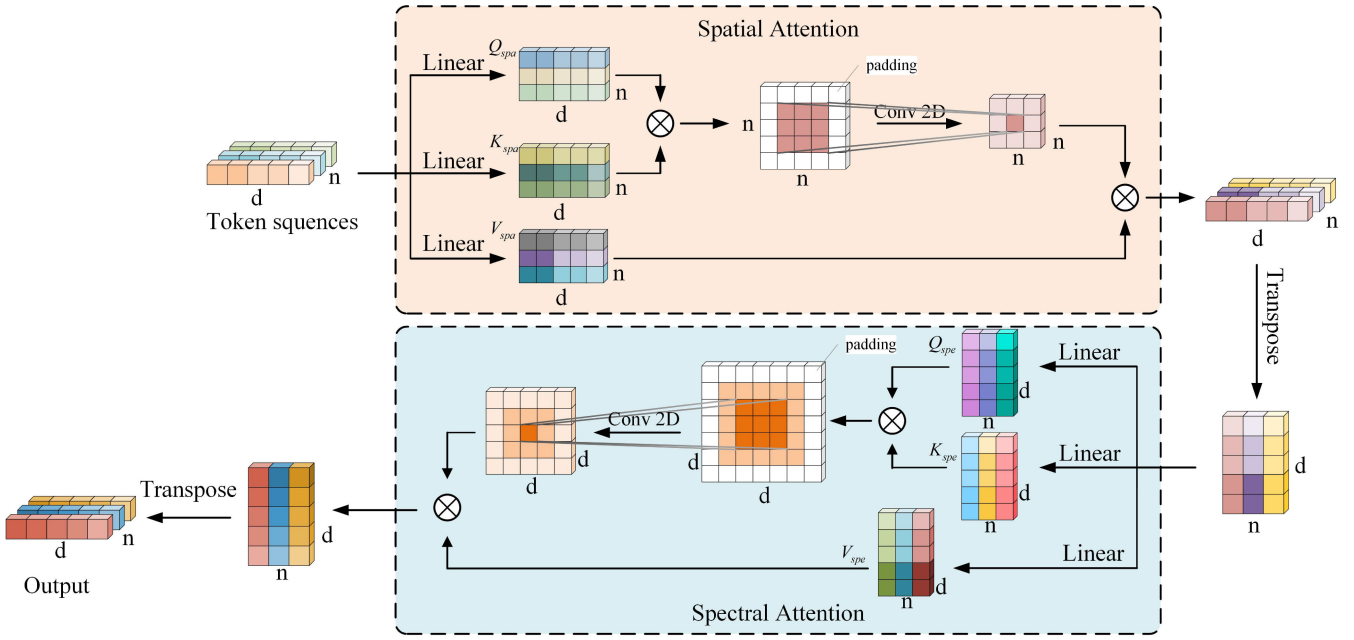


Fig. 5. Structure of the SSA module.

features belong to each class. Multiple classification results are obtained simultaneously. To effectively fuse the results of different granularity, an adaptive weighted fusion method is used to combine these results. Assuming that there are K branches and N categories to be distinguished and that B represents the classification result of each branch, the final classification result O can be expressed as

$$O = \sum_{i=1}^{K} W_i \times B_i \quad \text{s.t.} \sum_{i=1}^{K} W_i = 1 \tag{8}$$

where B_i denotes the classification of the ith granularity branch. W_i represents the weighting parameters of the ith granularity branch, which can be learned adaptively during the training process of HybridFormer.

The above is the entire structure of the HybridFormer model. In summary, the model can focus on the important positions and channels of in HSIs and ignore noisy backgrounds to adequately learn the global spatial–spectral information of multigranularity markers. It also explores further details from a local perspective, leveraging the abundant semantic information of the HSI.

III. Experiments

A. Datasets' Detail

Four publicly available HSI datasets were considered to verify the proposed HybridFormer model, including the Houston 2013 (HS) [38], WHU-Hi-HongHu (HongHu) [39], [40], Chikusei [41], and HyRANK [42] datasets.

The HS dataset was gathered by the National Center for Airborne Laser Mapping (NCALM) at the University of Houston, USA. It is composed of 349 × 1905 pixels with a spatial resolution of approximately 2.5 m and 144 spectral bands ranging between 0.38 and 1.05 μm. Fig. 6 demonstrates the false-color map, the ground-truth map, and the 15 different

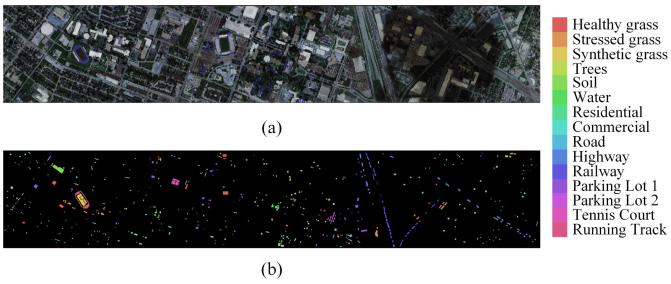


Fig. 6. Houston 2013 dataset: (a) false color and (b) ground truth.

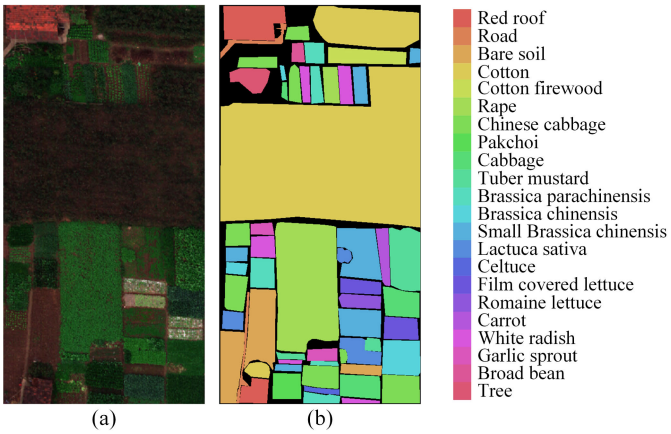


Fig. 7. HongHu dataset: (a) false color and (b) ground truth.

land-cover color class labels in the HS dataset. The names of the land-cover categories and the sample details are shown in Table II.

The HongHu dataset was acquired in an agricultural area with various crops in Hubei province, China. The images were obtained by a 17-mm focal length Headwall Nano-Hyperspec

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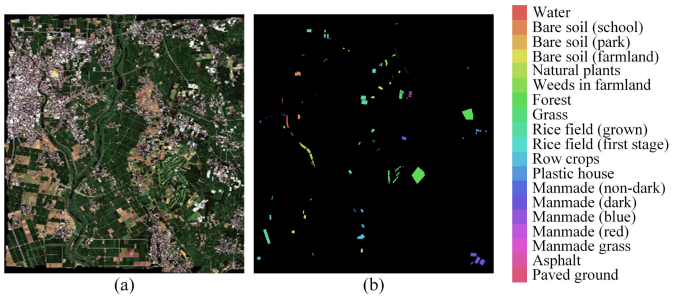


Fig. 8. Chikusei dataset: (a) false color and (b) ground truth.

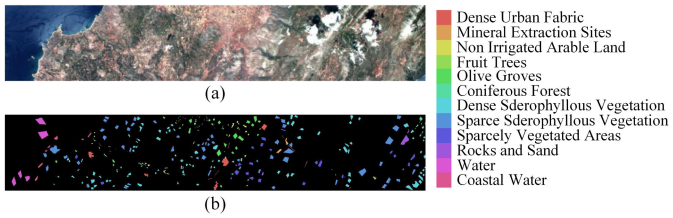


Fig. 9. HyRANK dataset: (a) false color and (b) ground truth.

TABLE II

DETAILS OF CLASSES IN THE HOUSTON DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train.	Test.
1	Healthy grass	13	1238
2	Stressed grass	13	1241
3	Synthetic grass	7	690
4	Trees	12	1232
5	Soil	12	1230
6	Water	3	322
7	Residential	13	1255
8	Commercial	12	1232
9	Road	13	1239
10	Highway	12	1215
11	Railway	12	1223
12	Parking Lot 1	12	1221
13	Parking Lot 2	5	464
14	Tennis Court	4	424
15	Running Track	7	653
	TOTAL	150	14879

imaging sensor fit on a DJI Matrice 600 Pro UAV platform with high spatial and spectral resolution. The HongHu dataset contains 940 × 475 pixels with a 0.043-m spatial resolution and 270 bands in a wavelength range of 0.4–1 μm. There are 22 categories in total, and the details can be found in Table III. The false-color composite images and ground-truth distribution color images are shown in Fig. 7(a) and (b), respectively.

The Chikusei dataset was collected using the Headwall's Hyperspec-VNIR-C imaging sensor in Chikusei, Ibaraki, Japan, on July 29, 2014. The dataset includes 128 spectral bands in the wavelength range of 0.363–1.018 μm and 2517 × 2335 pixels with a 2.5-m spatial resolution. The false-color images and label information are shown in Fig. 8. The 19 class names and the number of samples in each class are listed in Table IV.

The HyRANK dataset was also acquired by the EO-1 Hyperion sensor under the ISPRS Scientific Program framework. Only the Dioni scene was considered in this article. The Dioni image contains 250 × 1376 pixels and 176 spectral reflection bands. There are 12 categories of land cover, and the sample details are shown in Table V. False-color representations of the HyRANK and ground-truth image are displayed in Fig. 9.

TABLE III

DETAILS OF CLASSES IN THE WHU-HI-HONGHU DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train.	Test.
1	Red roof	140	13901
2	Road	35	3477
3	Bare soil	218	21603
4	Cotton	1632	161653
5	Cotton firewood	62	6156
6	Rape	446	44111
7	Chinese cabbage	241	23862
8	Pakchoi	41	4013
9	Cabbage	108	10711
10	Tuber mustard	124	12270
11	Brassica parachinensis	110	10905
12	Brassica chinensis	90	8864
13	Small Brassica chinensis	225	22282
14	Lactuca sativa	74	7282
15	Celtuce	10	992
16	Film covered lettuce	73	7189
17	Romaine lettuce	30	2980
18	Carrot	32	3185
19	White radish	87	8625
20	Garlic sprout	35	3451
21	Broad bean	13	1315
22	Tree	40	4000
	TOTAL	3866	382827

TABLE IV

DETAILS OF CLASSES IN THE CHIKUSEI DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train.	Test.
1	Water	28	2817
2	Bare soil (school)	29	2830
3	Bare soil (park)	3	283
4	Bare soil(farmland)	48	4804
5	Natural plants	43	4254
6	Weeds in farmland	11	1097
7	Forest	205	20311
8	Grass	65	6450
9	Rice field (grown)	134	13235
10	Rice field (first stage)	13	1255
11	Row crops	60	5901
12	Plastic house	22	2171
13	Manmade (non-dark)	12	1208
14	Manmade (dark)	77	7587
15	Manmade (blue)	4	427
16	Manmade (red)	2	220
17	Manmade grass	10	1030
18	Asphalt	8	793
19	Paved ground	1	144
	TOTAL	775	76817

B. Experimental Setting

All of the experiments were implemented on an Intel[1] Core[2] i7-10700F CPU @ 2.90 GHz with 16-GB RAM and an NVIDIA GeForce RTX 2060 SUPER 16-GB GPU. To reduce the experimental error, 1% of the labeled pixels were randomly selected as training samples. In addition, all of the

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TABLE V

DETAILS OF CLASSES IN THE HYRANK DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train	Test.
1	Dense Urban Fabric	13	1249
2	Mineral Extraction Sites	2	202
3	Non Irrigated Arable Land	6	608
4	Fruit Trees	1	149
5	Olive Groves	18	1750
6	Coniferous Forest	4	357
7	Dense Sderophyllous Vegetation	50	4985
8	Sparce Sderophyllous Vegetation	64	6310
9	Sparcely Vegetated Areas	17	1737
10	Rocks and Sand	5	487
11	Water	16	1596
12	Coastal Water	4	394
	TOTAL	200	19824

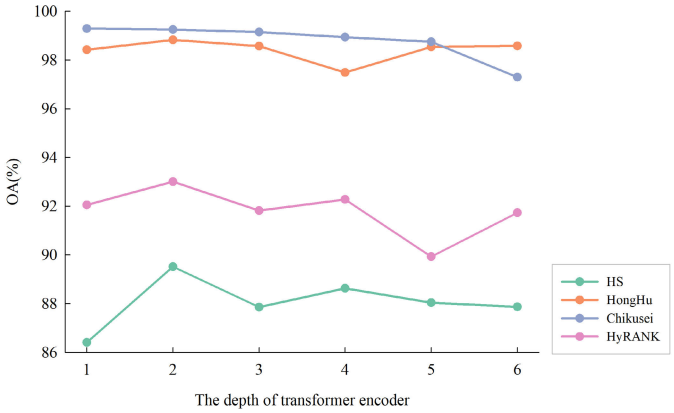


Fig. 10. OA comparison at different depths of the transformer.

experiments were conducted ten times, and the average results were recorded. Furthermore, to validate the effectiveness of the model, the overall accuracy (OA), average accuracy (AA), and Kappa coefficients (k) were used as evaluation metrics to assess the performance of the algorithm.

In addition, the settings of the transformer encoder are given as follows: the depth of each transformer encoder was set to 2 and the number of heads was set to 4. The subpatch size of the MDTE was set to 3 and 5. Furthermore, the learning rate was 0.0001, the number of training epochs was set to 200, and the batch size was 128. The input HSI was divided into patches of fixed sizes of 15 × 15.

C. Parameter Sensitivity Analysis

To conduct a comprehensive study of the proposed HybridFormer model, a number of key hyperparameters that are involved in the transformer encoder were initially analyzed. Subsequently, the depth of the transformer encoder, the number of attention heads, and the subpatch size of the MDTE were analyzed; these hyperparameters affect not only the accuracy of the HybridFormer but also its complexity. To analyze the effect of these parameters on the model, the other parameters were fixed and configured as presented in the experimental settings section.

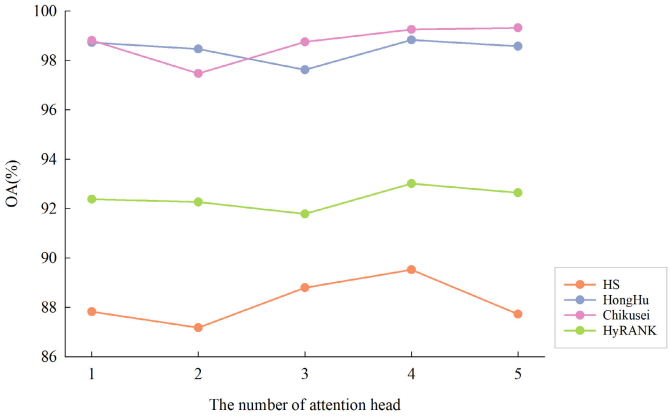


Fig. 11. OA comparison at different head numbers of the transformer.

TABLE VI

EFFECT OF DIFFERENT SUBPATCH SIZES ON OA FOR THE FOUR DATASETS (THE BEST TWO RESULTS ARE SHOWN IN RED AND BLUE)

Sub-patch Size				Dataset			
1	3	5	15	HS	HongHu	Chikusei	HyRANK
✓	✗	✗	✗	25.38	67.57	45.56	53.26
✗	✓	✗	✗	89.42	98.48	98.77	92.19
✗	✗	✓	✗	88.16	98.20	98.80	92.83
✗	✗	✗	✓	88.57	97.89	98.92	**93.43**
✓	✓	✗	✗	78.65	95.76	93.87	89.78
✓	✗	✓	✗	77.73	95.57	92.84	90.41
✓	✗	✗	✓	73.17	94.88	91.39	82.63
✗	✓	✓	✗	**89.52**	**98.83**	**99.25**	**93.01**
✗	✓	✗	✓	**90.11**	98.61	97.95	92.59
✗	✗	✓	✓	88.81	98.49	98.41	92.92
✗	✓	✓	✓	89.43	**98.66**	**99.07**	91.07
✓	✗	✓	✓	80.88	96.59	97.35	89.08
✓	✓	✗	✓	78.74	97.07	94.26	90.70
✓	✓	✓	✗	80.66	97.56	96.33	90.71
✓	✓	✓	✓	84.18	96.97	97.73	89.96

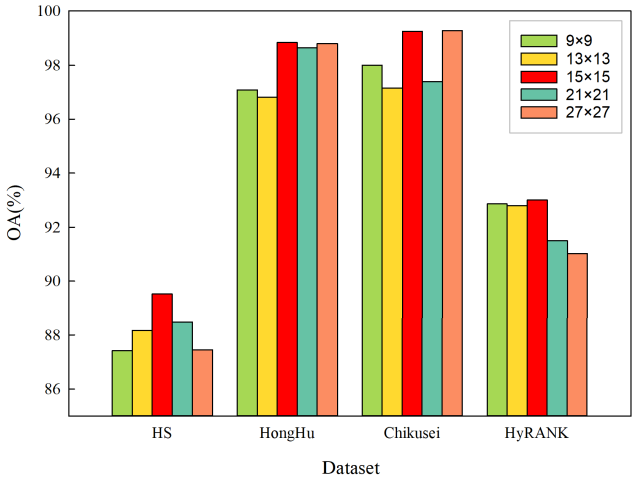


Fig. 12. OA comparison at different patch sizes.

1) Transformer Parameter Analysis: First, only the depth of the transformer encoder was changed from 1 to 6 for comparison, and the other parameters were fixed. The OA

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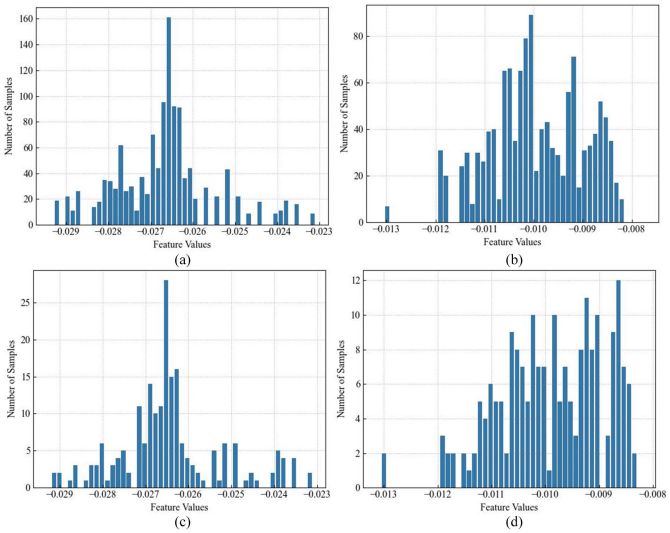


Fig. 13. Feature distribution of different subpatch sizes on the HyRANK dataset. (a) Dense Urban Fabric with 3 × 3 subpatch size. (b) Dense Urban Fabric with 5 × 5 subpatch size. (c) Mineral Extraction Sites with 3 × 3 subpatch size. (d) Mineral Extraction Sites with 5 × 5 subpatch size.

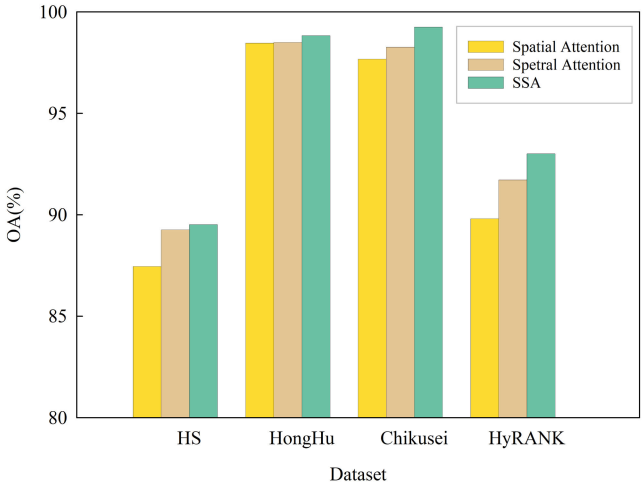


Fig. 14. Ablation study results on the SSA module.

results for each dataset are shown in Fig. 10, and it is obvious that the optimal depth of the model is 2. To search for the optimal number of attention heads, we set the number of heads from 1 to 5, respectively, for the experiment. As shown in Fig. 11, the best number of attention heads is 4 considering the four datasets. Ultimately, 2 and 4 were chosen as the depth and the number of attention heads of the HybridFormer model, respectively.

2) Effects of Different Patch Sizes: In addition, the effect of patch size on the classification result was further evaluated. Herein, we selected the optimal patch size from 9 × 9, 11 × 11, 15 × 15, 23 × 23, and 27 × 27. To control for variables, the subpatch size was set to 3 × 3 and 5 × 5. Fig. 12 presents the average OA of ten times the results for different patch sizes on the four datasets. As shown in Fig. 12, the OA increases and then decreases with increasing patch size on the HS dataset due to the fact that larger patches increase information, while larger patches introduce noise. The other three datasets do

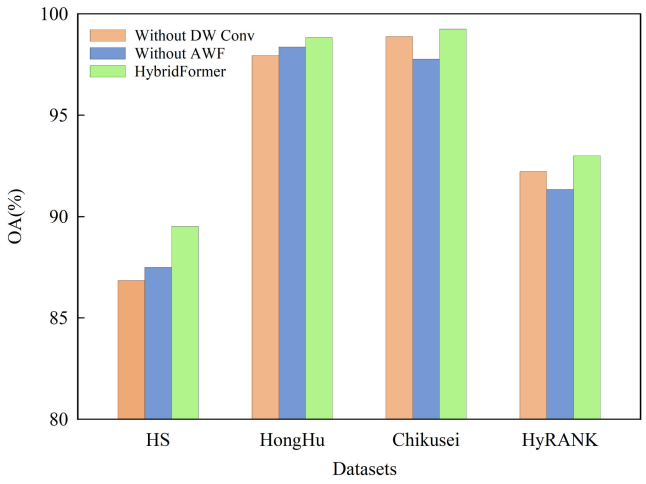


Fig. 15. Ablation study results on the detail elements.

TABLE VII

ABLATION STUDY RESULTS OF THE PROPOSED COMPONENTS ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Component			Dataset			
CNN feature extractor	MDTE	SSA	HS	HongHu	Chikusei	HyRANK
✗	✗	✗	78.39	96.3	98.10	89.22
✓	✗	✗	85.79	98.65	98.50	92.85
✗	✓	✗	81.51	97.52	98.36	89.78
✗	✗	✓	81.82	96.8	98.94	89.91
✓	✓	✗	87.43	98.69	98.51	92.1
✗	✓	✓	81.95	96.64	98.64	89.49
✓	✗	✓	89.09	98.67	98.59	**93.17**
✓	✓	✓	**89.52**	**98.83**	**99.25**	93.01

not show a regular trend, possibly due to noise, unbalanced sample sizes in each category, and the degree of dispersion in the sample distribution. However, it is obvious from Fig. 13 that the OA of 15 × 15 patch size has a high OA on all datasets. Therefore, we chose 15 × 15 as the optimal size for the HybridFormer.

3) Effects of Different Subpatch Sizes: To verify the effectiveness of the multigranularity embedding strategy, we tested the effect of subpatch size on the model. It is worth mentioning that the size of the input patch selected in this study is 15 × 15, so the subpatches can only be partitioned into four sizes 1, 3, 5, and 15. A randomly selected 1% labeled training sample was trained, and its ten times average OA is listed in Table VI. The results show that, when the branch with a subpatch size of 1 × 1 is selected, the accuracy rate decreases. Overall, the highest accuracy was obtained when the combination of 3 × 3 and 5 × 5 was chosen. Compared to using a single branch of subpatch size 3 × 3 or 5 × 5 alone, the combination of 3 × 3 and 5 × 5 improved the OA result on all datasets.

To visualize the discrepancy of the different granularity of the inputs, subpatches of different sizes for the Dense Urban Fabric and Mineral Extraction Sites in the HyRANK dataset were plotted, as shown in Fig. 13. The horizontal coordinate "feature values" represents the average value of the feature map of one sample, and the vertical coordinate "number of samples" indicates the frequency of the feature values falling within a specified range. Fig. 13(a)–(d) represents the distribution of features for different subpatch sizes samples

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TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET (THE BEST RESULT ARE SHOWN IN BOLD)

Class	SVM	KNN	2D-CNN	HybridSN	SSRN	DBDA	RSSAN	FDSSC	ViT	SpectralFormer	RvT	HiT	HybridFormer
1	90.46 (±2.67)	**91.24 (±2.17)**	89.34 (±1.02)	88.18 (±2.28)	88.81 (±3.65)	85.30 (±1.59)	86.60 (±3.07)	81.20 (±3.36)	84.62 (±5.08)	86.88 (±2.44)	87.87 (±1.54)	85.60 (±2.58)	89.60 (±2.09)
2	**90.96 (±5.00)**	90.46 (±1.43)	89.44 (±1.27)	88.60 (±1.96)	88.39 (±4.99)	84.47 (±2.44)	79.24 (±3.41)	81.62 (±1.35)	80.63 (±4.46)	80.06 (±3.82)	83.89 (±4.69)	87.78 (±3.00)	89.18 (±3.68)
3	96.44 (±1.23)	93.68 (±1.23)	98.54 (±0.36)	87.26 (±12.21)	99.12 (±0.81)	97.79 (±1.13)	95.72 (±3.70)	97.91 (±0.92)	88.86 (±9.02)	73.85 (±8.13)	91.84 (±3.58)	95.75 (±2.10)	**99.20 (±0.77)**
4	93.06 (±2.08)	92.90 (±1.86)	94.14 (±0.79)	92.08 (±2.76)	92.65 (±7.02)	88.55 (±3.18)	85.61 (±6.64)	79.01 (±4.16)	86.47 (±5.73)	88.98 (±4.99)	89.15 (±2.63)	89.79 (±3.97)	**95.14 (±2.82)**
5	91.00 (±3.97)	87.36 (±4.73)	94.58 (±3.10)	95.02 (±1.67)	95.96 (±3.18)	**97.83 (±1.85)**	92.14 (±2.97)	97.29 (±0.72)	96.66 (±2.09)	79.24 (±4.82)	97.58 (±1.39)	95.83 (±1.95)	97.34 (±1.15)
6	85.40 (±3.52)	**89.88 (±2.87)**	81.58 (±5.54)	66.98 (±9.76)	82.04 (±9.13)	87.66 (±3.87)	60.88 (±13.49)	84.03 (±3.36)	77.96 (±9.17)	42.73 (±11.99)	74.18 (±10.18)	80.96 (±8.69)	87.87 (±3.76)
7	71.84 (±8.35)	71.14 (±2.38)	86.64 (±2.22)	79.96 (±3.00)	85.71 (±6.09)	85.86 (±3.65)	79.60 (±4.35)	70.90 (±10.66)	81.02 (±3.75)	80.50 (±2.41)	83.94 (±2.30)	83.83 (±5.42)	**89.98 (±3.09)**
8	58.98 (±7.00)	57.80 (±9.88)	70.48 (±5.58)	57.28 (±9.09)	67.67 (±2.72)	76.58 (±7.81)	59.13 (±5.84)	56.66 (±12.73)	70.10 (±7.41)	49.50 (±9.22)	74.78 (±5.12)	71.80 (±5.23)	**80.03 (±5.92)**
9	66.86 (±5.35)	64.58 (±2.84)	**86.88 (±2.01)**	83.64 (±3.46)	79.24 (±9.56)	82.33 (±2.14)	75.02 (±4.40)	68.87 (±12.20)	76.43 (±4.38)	68.81 (±3.01)	77.22 (±5.93)	81.08 (±4.56)	83.71 (±3.06)
10	67.54 (±4.79)	60.98 (±5.28)	80.38 (±4.64)	72.48 (±6.21)	76.58 (±3.13)	79.15 (±7.06)	73.94 (±8.77)	70.75 (±5.06)	68.19 (±8.47)	55.88 (±6.48)	70.69 (±7.31)	77.88 (±4.31)	**85.73** (±3.01)
11	58.22 (±6.98)	60.40 (±4.84)	83.44 (±4.80)	71.68 (±2.17)	82.09 (±5.43)	83.24 (±7.02)	67.80 (±7.66)	72.85 (±4.52)	72.28 (±3.75)	50.74 (±6.60)	71.92 (±7.68)	79.23 (±5.92)	**88.39 (±4.35)**
12	54.68 (±6.69)	52.80 (±3.19)	81.96 (±2.83)	67.12 (±5.15)	80.01 (±6.04)	79.88 (±5.11)	74.77 (±4.63)	55.70 (±9.99)	72.61 (±9.86)	49.30 (±5.45)	77.58 (±7.52)	83.37 (±5.11)	**86.30 (±3.05)**
13	18.34 (±10.01)	17.64 (±5.71)	**84.72 (±2.24)**	88.70 (±3.79)	81.67 (±8.61)	77.25 (±6.84)	64.24 (±7.81)	51.55 (±6.10)	41.75 (±15.98)	12.28 (±6.74)	78.79 (±6.44)	71.00 (±10.90)	80.14 (±8.60)
14	85.90 (±6.14)	86.74 (±5.95)	89.58 (±6.07)	91.18 (±5.33)	93.88 (±5.42)	**98.63 (±1.12)**	67.23 (±10.44)	93.57 (±3.05)	97.76 (±1.74)	64.72 (±3.71)	94.78 (±4.56)	95.78 (±3.63)	98.32 (±2.36)
15	**98.58 (±0.60)**	95.78 (±2.62)	96.96 (±1.42)	96.50 (±2.75)	97.98 (±0.71)	95.46 (±1.89)	83.13 (±9.33)	97.18 (±1.20)	92.55 (±2.14)	70.62 (±11.05)	94.80 (±2.51)	91.69 (±4.67)	97.43 (±0.77)
OA	75.82 (±1.64)	74.41 (±2.02)	87.03 (±0.94)	81.24 (±1.49)	86.02 (±2.80)	85.83 (±1.74)	77.82 (±3.42)	76.18 (±1.74)	79.51 (±1.99)	67.80 (±2.80)	82.79 (±2.38)	84.37 (±1.33)	**89.52 (±0.53)**
AA	75.64 (±1.53)	74.03 (±1.88)	88.37 (±0.46)	82.07 (±1.39)	86.57 (±3.24)	86.94 (±1.71)	75.51 (±3.49)	79.29 (±1.07)	78.92 (±1.92)	63.92 (±2.69)	82.58 (±2.28)	84.76 (±1.80)	**89.81 (±0.77)**
k×100	73.84 (±1.77)	76.31 (±7.45)	85.99 (±1.01)	79.94 (±1.45)	84.88 (±3.04)	84.68 (±1.88)	75.99 (±3.69)	74.32 (±1.85)	77.84 (±2.15)	65.12 (±3.04)	81.39 (±2.57)	83.11 (±1.44)	**88.67 (±0.57)**

in the Dense Urban Fabric and Mineral Extraction Sites, respectively. The comparison of Fig. 13(a)–(d) shows that the feature distribution of different size samples is highly variable for the same class. Moreover, the distribution is more uniform and concentrated for inputs with larger granularity, while the distribution is broader for inputs with smaller granularity. Therefore, the use of multigranularity features can provide richer information to the model and further improve the classification performance.

D. Ablation Study

1) Ablation Experiments on the Main Modules: To analyze the contribution of the detailed components in the HybridFormer model, ablation studies with different combinations of elements were conducted on the four datasets. The entire HybridFormer model comprises three main components: the CNN feature extractor, MDTE, and SSA. Table VII records eight combinations of the three structures; the effect of the different components on the model was analyzed from the perspective of OA. In this case, the absence of MDTE means that a single granularity of 5 was used, and the absence of SSA indicates that only spatial attention was used.

Overall, the model without all three components has the worst classification accuracy on all of the datasets. The addition of the CNN feature extraction module alone resulted in accuracy increases of 7.4%, 2.35%, 0.4%, and 3.63% on each dataset, which had the greatest impact on the accuracy in comparison. It demonstrates that using the extracted shallow semantic feature as the token of the transformer allows for better global modeling. It is worth noting that the addition of any one component alone improves the accuracy over without them, which illustrates that each of the proposed components makes some contribution to HSI classification. In addition, the model performs best with all three components, with 11.13%, 2.53%, 1.15%, and 3.79% improvements in terms of the OA compared to running the model without them. Furthermore, the combination of any two components is more precise than one alone, except for the case in which only without CNN feature extractor was excluded.

2) Ablation Study on the SSA Module: To further validate the effectiveness of the SSA module, we compared the influence of the three attention structures on the classification performance, including spatial attention, spectral attention, and the proposed SSA. The rest of the network structures were kept consistent. Fig. 14 shows the classification results of the three different attention modules in terms of the OA, AA, and k metrics. It can be seen that the model's accuracy of the model based on spectral attention is higher than the model based on spatial attention. The result indicates that spectral features play a very important role in the classification task. Moreover, the accuracy of the SSA model is higher than that of the other two models on most datasets, which demonstrates that the SSA module further improves the hyperspectral classification performance by focusing the model on discriminative positions and channels.

3) Ablation Study on the Detail Elements: To validate the effectiveness of the proposed model, we also conducted

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TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE WHU-HI-HONGHU DATASET (THE BEST RESULT ARE SHOWN IN BOLD)

Class	SVM	KNN	2D-CNN	HybridSN	SSRN	DBDA	RSSAN	FDSSC	ViT	SpectralFormer	RvT	HiT	HybridFormer
1	92.56 (±0.45)	90.16 (±0.80)	98.26 (±0.16)	95.20 (±2.40)	94.61 (±7.08)	98.38 (±0.59)	96.98 (±0.62)	98.32 (±0.62)	98.62 (±0.43)	98.06 (±0.44)	97.94 (±0.24)	98.77 (±0.25)	**99.01 (±0.39)**
2	71.00 (±2.24)	69.20 (±0.97)	91.44 (±1.88)	82.92 (±5.65)	88.58 (±5.24)	87.39 (±2.14)	86.18 (±3.01)	82.07 (±3.38)	90.89 (±2.48)	89.79 (±2.70)	87.74 (±3.35)	90.97 (±1.54)	**93.47 (±1.04)**
3	84.16 (±0.44)	84.28 (±0.39)	96.32 (±1.41)	95.24 (±1.85)	95.64 (±4.08)	97.81 (±0.43)	94.27 (±1.27)	96.71 (±0.79)	96.24 (±1.93)	95.15 (±0.75)	96.17 (±0.44)	96.82 (±0.52)	**98.39 (±1.16)**
4	96.32 (±0.07)	94.16 (±0.12)	98.42 (±0.59)	97.74 (±2.82)	97.15 (±4.77)	99.61 (±0.10)	98.97 (±0.59)	99.53 (±0.12)	99.51 (±0.07)	99.56 (±0.13)	99.56 (±0.07)	99.58 (±0.03)	**99.70 (±0.02)**
5	52.08 (±1.17)	28.44 (±2.39)	83.30 (±6.79)	84.18 (±7.18)	92.22 (±6.58)	94.38 (±2.32)	87.89 (±4.56)	92.52 (±2.51)	92.53 (±1.93)	91.64 (±2.04)	93.59 (±1.06)	94.69 (±1.06)	**96.67 (±0.79)**
6	88.02 (±0.22)	82.72 (±0.37)	99.00 (±0.11)	97.86 (±1.35)	97.01 (±4.14)	99.48 (±0.17)	97.04 (±0.54)	99.39 (±0.23)	98.98 (±0.32)	96.63 (±0.40)	98.58 (±0.49)	98.98 (±0.15)	**99.56 (±0.18)**
7	71.42 (±0.96)	65.70 (±0.45)	95.38 (±0.61)	91.50 (±3.49)	95.81 (±1.54)	97.73 (±0.21)	90.32 (±1.52)	97.78 (±0.64)	94.68 (±4.62)	89.67 (±1.48)	95.12 (±1.54)	96.30 (±0.24)	**98.41 (±0.39)**
8	20.64 (±2.70)	8.78 (±3.06)	83.32 (±3.36)	74.32 (±16.10)	88.99 (±6.53)	94.37 (±1.99)	61.81 (±2.17)	97.39 (±1.54)	87.37 (±3.16)	62.42 (±2.88)	79.47 (±6.46)	86.48 (±2.22)	**95.44 (±2.40)**
9	94.20 (±0.38)	90.84 (±0.80)	99.16 (±0.35)	95.34 (±3.34)	98.24 (±2.07)	99.16 (±0.57)	97.94 (±0.70)	99.03 (±0.38)	98.67 (±0.34)	98.19 (±0.28)	98.87 (±0.36)	99.32 (±0.13)	**99.53 (±0.21)**
10	58.80 (±1.39)	36.30 (±2.78)	97.10 (±0.34)	91.22 (±4.91)	96.80 (±0.79)	97.29 (±0.48)	89.49 (±2.14)	97.12 (±2.11)	95.09 (±2.42)	85.30 (±2.13)	94.87 (±0.85)	97.29 (±0.34)	**98.01 (±0.31)**
11	48.92 (±1.19)	31.48 (±1.96)	95.66 (±0.52)	84.24 (±8.21)	90.24 (±11.84)	97.65 (±0.64)	84.29 (±3.18)	97.76 (±0.58)	95.33 (±1.59)	84.99 (±1.44)	94.14 (±1.07)	95.42 (±1.11)	**98.07 (±0.43)**
12	53.50 (±1.57)	51.48 (±1.23)	91.06 (±1.06)	84.52 (±7.52)	94.48 (±1.97)	96.22 (±1.52)	82.93 (±2.43)	94.03 (±5.89)	90.36 (±5.10)	84.25 (±3.96)	91.13 (±1.02)	92.31 (±0.99)	**96.75 (±0.94)**
13	62.52 (±1.41)	60.78 (±1.00)	91.96 (±3.98)	90.74 (±3.67)	94.06 (±3.73)	97.15 (±0.75)	86.25 (±1.55)	96.84 (±1.76)	93.94 (±2.00)	87.08 (±0.36)	92.48 (±1.47)	95.07 (±0.73)	**97.74 (±0.47)**
14	71.72 (±1.12)	63.08 (±0.98)	97.66 (±0.23)	94.44 (±2.61)	97.21 (±0.62)	96.90 (±1.00)	91.07 (±2.71)	97.51 (±0.77)	96.83 (±0.59)	91.00 (±0.42)	96.21 (±0.73)	96.43 (±0.35)	**98.37 (±0.72)**
15	52.46 (±4.17)	27.16 (±6.01)	93.46 (±4.03)	64.82 (±26.54)	91.07 (±6.60)	93.55 (±2.28)	88.96 (±2.01)	93.72 (±1.20)	94.25 (±2.40)	84.46 (±3.88)	90.73 (±4.58)	92.88 (±1.04)	**95.96 (±1.41)**
16	84.54 (±0.76)	82.04 (±1.02)	98.40 (±0.62)	91.84 (±11.59)	98.83 (±0.81)	98.18 (±0.53)	90.38 (±2.07)	98.62 (±2.13)	95.60 (±6.48)	95.48 (±1.19)	96.60 (±2.22)	97.62 (±1.04)	**99.60 (±0.20)**
17	66.78 (±2.19)	62.58 (±3.52)	96.76 (±1.43)	85.86 (±10.45)	94.98 (±4.42)	96.40 (±1.11)	81.09 (±5.43)	97.40 (±4.10)	92.37 (±8.36)	89.91 (±3.19)	92.95 (±4.99)	91.86 (±5.01)	**98.81 (±0.84)**
18	48.66 (±2.60)	29.08 (±2.09)	94.56 (±1.29)	80.62 (±18.56)	96.01 (±1.01)	94.36 (±1.55)	87.79 (±5.20)	93.67 (±6.04)	88.35 (±16.20)	80.33 (±4.75)	89.31 (±1.26)	94.20 (±1.60)	**96.49 (±1.20)**
19	74.32 (±0.87)	64.36 (±0.97)	96.10 (±0.51)	90.64 (±5.32)	96.27 (±2.12)	97.13 (±0.52)	93.24 (±1.12)	97.29 (±0.80)	95.03 (±1.28)	91.32 (±1.51)	95.28 (±0.58)	96.57 (±0.95)	**97.78 (±0.56)**
20	64.86 (±2.67)	50.16 (±1.28)	94.12 (±1.59)	85.34 (±6.65)	95.36 (±2.44)	97.12 (±0.88)	91.25 (±3.07)	97.32 (±1.31)	94.80 (±1.64)	89.52 (±1.35)	95.92 (±0.92)	96.39 (±0.90)	**97.67 (±1.13)**
21	13.96 (±1.40)	8.38 (±2.33)	71.08 (±15.10)	81.94 (±12.00)	91.34 (±4.72)	96.15 (±3.09)	54.25 (±4.52)	**97.58 (±2.03)**	77.35 (±16.02)	70.51 (±5.19)	69.45 (±2.95)	71.32 (±4.97)	94.67 (±3.33)
22	46.32 (±3.32)	25.44 (±5.29)	83.86 (±8.07)	80.28 (±11.84)	88.88 (±13.42)	97.71 (±1.02)	85.43 (±11.25)	**98.20 (±0.63)**	96.17 (±2.43)	95.50 (±1.19)	96.38 (±1.11)	96.88 (±0.65)	97.99 (±0.61)
OA	83.17 (±0.28)	73.33 (±0.20)	96.54 (±0.62)	93.84 (±4.14)	95.89 (±4.43)	98.36 (±0.24)	94.11 (±0.96)	98.15 (±0.45)	97.08 (±1.28)	94.80 (±0.23)	96.95 (±0.49)	97.69 (±0.11)	**98.83 (±0.09)**
AA	62.59 (±0.26)	52.45 (±0.18)	95.60 (±0.57)	88.91 (±5.92)	95.69 (±2.46)	97.07 (±0.70)	88.42 (±0.91)	97.48 (±1.03)	94.62 (±1.44)	88.05 (±0.32)	92.98 (±1.14)	94.05 (±0.18)	**97.87 (±0.30)**
K ×100	78.59 (±0.31)	79.37 (±0.25)	95.67 (±0.76)	92.32 (±5.05)	94.94 (±5.32)	97.93 (±0.31)	92.58 (±1.19)	97.67 (±0.57)	96.2 (±1.61)	93.42 (±0.29)	96.15 (±0.62)	97.09 (±0.14)	**98.53 (±0.11)**

ablation experiments on several detailed elements of the network, including depthwise convolution in the MTG (Without DW Conv) and the adaptive weighted fusion method (Without AWF). As shown in Fig. 15, the model without depthwise convolution and without adaptive weighted fusion has lower accuracy than the HybridFormer on all four datasets, which indicates that depthwise convolution in the MTG can separate the spatial and spectral information and allow the model to focus on differentiated locations and spectral. In addition, the proposed adaptive weighted fusion enables the network to fully fuse feature information of different granularities, which further improves classification accuracy.

E. Comparison Experiments

To validate the advantages of the HybridFormer model, we implemented 12 representative algorithms as comparison methods, including the SVM, KNN, 2-D-CNN [43], HybridSN [12], SSRN [44], DBDA [45], RSSAN [46], FDSSC [47], SpectralFormer [48], ViT [34], RvT [49], and HiT [50]. For the SVM model, we used the RBF kernel function and set three different kernel function coefficients of 1, 10, and 1000 and four different penalty coefficients of 1, 10, 100, and 1000. In addition, grid search and K-fold cross-validation methods were used to optimize and evaluate the model, respectively.

For the KNN model, five different numbers of neighbors, 1, 3, 5, 10, and 20, were set. In addition, like the SVM model, grid search and K-fold cross-validation methods were also used to optimize and evaluate the KNN model. For the 2-D-CNN model, it was composed of three 2-D convolutional blocks, a maxpooling layer, and two linear layers. Each 2-D convolutional block consisted of a 2-D convolution, a BN layer, and an ReLU activation function. The first 2-D convolutional layer contained 64 filters with a size of 1, followed by a maxpooling layer with a size of 2. The second and third 2-D convolutional layers contained 128 and 256 filters with a size of 3, respectively. For the ViT method, the network was composed of a transformer and two linear layers. The depth of the transformer and the number of heads were set to 2 and 4, respectively. The rest of the structures were consistent

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TABLE X

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE CHIKUSEI DATASET (THE BEST RESULT ARE SHOWN IN BOLD)

Class	SVM	KNN	2D-CNN	HybridSN	SSRN	DBDA	RSSAN	FDSSC	ViT	SpectralFormer	RvT	HiT	HybridFormer
1	99.22 (±0.87)	95.84 (±1.94)	97.48 (±1.16)	94.98 (±3.27)	97.65 (±3.61)	**99.71 (±0.39)**	90.49 (±3.32)	98.95 (±1.63)	98.26 (±1.65)	94.47 (±2.14)	92.81 (±9.30)	99.37 (±0.87)	99.62 (±0.24)
2	97.10 (±0.57)	96.20 (±1.02)	99.52 (±0.39)	97.42 (±0.84)	98.91 (±0.95)	99.25 (±0.66)	95.20 (±5.47)	99.55 (±0.39)	99.41 (±0.36)	98.60 (±0.36)	98.44 (±1.64)	99.69 (±0.17)	**99.83 (±0.21)**
3	32.82 (±14.41)	32.36 (±5.28)	49.58 (±10.70)	35.34 (±11.07)	60.00 (±16.82)	**78.25 (±12.48)**	46.72 (±10.93)	58.60 (±9.72)	55.79 (±7.78)	33.57 (±3.96)	52.46 (±9.21)	49.40 (±8.08)	71.78 (±9.40)
4	92.82 (±4.26)	93.78 (±0.61)	97.00 (±1.06)	89.30 (±6.14)	88.93 (±8.16)	97.26 (±1.90)	91.54 (±4.19)	96.04 (±2.04)	96.25 (±1.19)	94.42 (±0.66)	93.96 (±4.05)	96.70 (±0.42)	**97.33 (±1.61)**
5	97.46 (±0.69)	97.72 (±0.38)	99.16 (±1.14)	99.46 (±0.89)	99.92 (±0.13)	99.96 (±0.07)	98.78 (±1.50)	**99.99 (±0.01)**	99.97 (±0.06)	98.85 (±0.68)	97.00 (±5.91)	99.98 (±0.01)	99.96 (±0.07)
6	88.76 (±1.98)	79.04 (±5.10)	70.36 (±23.52)	97.16 (±2.50)	79.79 (±24.74)	96.02 (±4.41)	89.86 (±5.79)	96.55 (±2.07)	98.06 (±1.46)	92.85 (±2.61)	88.06 (±21.66)	**98.35 (±2.02)**	97.12 (±2.23)
7	99.40 (±0.23)	98.04 (±0.22)	95.24 (±3.51)	99.66 (±0.25)	97.70 (±4.36)	99.99 (±0.03)	99.67 (±0.26)	99.99 (±0.01)	99.99 (±0.03)	99.94 (±0.02)	99.54 (±0.91)	**100.00 (±0.01)**	99.99 (±0.03)
8	99.02 (±0.12)	97.44 (±0.55)	99.04 (±1.06)	98.94 (±0.72)	99.15 (±0.86)	99.75 (±0.10)	98.31 (±0.80)	99.80 (±0.05)	99.34 (±0.45)	98.95 (±0.22)	98.80 (±1.86)	**99.81 (±2.02)**	99.65 (±0.16)
9	99.58 (±0.17)	97.86 (±0.17)	99.46 (±0.39)	99.36 (±0.50)	99.04 (±1.08)	99.58 (±0.47)	99.56 (±0.32)	99.62 (±0.22)	99.82 (±0.10)	99.53 (±0.27)	97.50 (±4.69)	**99.86 (±0.17)**	99.81 (±0.14)
10	99.60 (±0.26)	99.28 (±0.68)	93.28 (±2.87)	97.22 (±2.41)	99.60 (±0.42)	99.63 (±0.73)	95.14 (±1.24)	98.58 (±2.38)	98.38 (±1.77)	93.73 (±2.87)	83.15 (±28.80)	**100.00 (±0.00)**	**100.00 (±0.00)**
11	98.66 (±0.63)	96.86 (±0.27)	99.50 (±0.25)	97.74 (±1.72)	96.30 (±5.27)	99.52 (±0.54)	96.51 (±1.55)	99.83 (±0.32)	**99.00 (±0.80)**	97.11 (±0.89)	98.10 (±2.75)	**99.90 (±0.09)**	99.49 (±0.61)
12	92.80 (±1.46)	90.16 (±1.89)	97.16 (±1.23)	93.74 (±3.78)	95.53 (±1.02)	96.38 (±3.60)	89.49 (±2.89)	**97.64 (±1.74)**	95.94 (±1.60)	94.80 (±0.77)	94.58 (±4.70)	96.90 (±1.77)	97.24 (±2.65)
13	95.82 (±1.61)	95.60 (±1.55)	**98.18 (±0.60)**	97.12 (±1.14)	97.03 (±1.25)	96.68 (±3.99)	93.44 (±2.51)	98.10 (±0.53)	97.43 (±1.06)	96.53 (±0.86)	96.83 (±3.71)	97.74 (±0.66)	96.69 (±2.53)
14	99.10 (±0.40)	98.08 (±0.73)	99.36 (±0.50)	98.40 (±0.90)	98.60 (±1.48)	99.83 (±0.19)	96.79 (±1.53)	99.61 (±0.19)	99.25 (±0.23)	98.19 (±0.46)	98.18 (±1.74)	99.55 (±0.31)	**99.81 (±0.09)**
15	89.16 (±16.12)	99.50 (±0.28)	93.82 (±6.29)	96.40 (±2.79)	97.73 (±1.55)	95.75 (±8.72)	82.24 (±12.46)	**99.81 (±0.09)**	95.71 (±4.91)	95.50 (±3.23)	93.59 (±5.38)	92.14 (±9.13)	93.98 (±9.62)
16	93.04 (±4.23)	91.02 (±1.77)	93.56 (±5.29)	88.18 (±4.01)	88.61 (±8.63)	91.86 (±6.62)	64.16 (±25.45)	85.59 (±0.36)	93.11 (±5.95)	87.24 (±2.63)	78.22 (±19.78)	**97.40 (±3.01)**	90.39 (±6.20)
17	95.26 (±1.35)	94.90 (±1.50)	96.74 (±3.42)	94.66 (±2.54)	98.71 (±0.95)	98.73 (±0.96)	90.96 (±4.47)	**99.55 (±0.89)**	97.60 (±1.17)	89.83 (±1.60)	95.73 (±1.77)	97.48 (±0.92)	98.54 (±0.84)
18	89.32 (±3.59)	82.74 (±6.45)	84.64 (±2.92)	84.66 (±5.76)	85.97 (±4.65)	91.89 (±4.09)	78.11 (±4.68)	88.72 (±4.83)	92.34 (±3.93)	84.22 (±6.20)	88.55 (±5.22)	**95.66 (±1.86)**	95.63 (±2.54)
19	47.48 (±21.92)	36.94 (±16.01)	23.16 (±7.82)	73.32 (±12.60)	75.67 (±18.67)	91.34 (±22.28)	16.85 (±11.70)	**95.40 (±4.62)**	54.11 (±26.69)	38.49 (±8.04)	57.13 (±12.35)	81.85 (±11.46)	90.33 (±12.84)
OA	97.71 (±0.31)	96.99 (±0.98)	96.32 (±1.86)	97.29 (±1.23)	96.53 (±2.47)	99.14 (±0.41)	96.54 (±0.94)	98.98 (±0.20)	98.86 (±0.20)	97.70 (±0.40)	96.88 (±3.79)	99.18 (±0.16)	**99.25 (±0.23)**
AA	89.50 (±3.35)	87.06 (±0.60)	92.63 (±0.64)	91.98 (±1.02)	95.40 (±1.79)	**97.24 (±2.09)**	84.28 (±2.13)	96.79 (±1.58)	92.36 (±1.59)	87.59 (±0.76)	89.98 (±4.25)	94.12 (±2.50)	96.28 (±1.09)
k×100	97.36 (±0.35)	95.83 (±0.30)	95.60 (±2.10)	96.87 (±1.42)	96.02 (±2.81)	99.01 (±0.47)	96.01 (±1.08)	98.83 (±0.23)	98.68 (±0.23)	97.35 (±0.46)	96.41 (±4.36)	99.06 (±0.18)	**99.14 (±0.27)**

TABLE XI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HYRANK DATASET (THE BEST RESULT ARE SHOWN IN BOLD)

Class	SVM	KNN	2D-CNN	HybridSN	SSRN	DBDA	RSSAN	FDSSC	ViT	SpectralFormer	RvT	HiT	HybridFormer
1	60.78 (±4.32)	47.14 (±4.96)	81.46 (±4.36)	76.10 (±8.94)	86.71 (±1.06)	**87.28 (±2.42)**	71.15 (±6.32)	86.02 (±3.10)	80.40 (±11.12)	24.16 (±4.66)	82.25 (±4.42)	86.28 (±1.03)	86.62 (±2.19)
2	56.30 (±17.67)	56.48 (±10.27)	74.30 (±12.03)	73.90 (±11.07)	83.56 (±12.01)	**93.14 (±8.26)**	30.26 (±9.59)	69.19 (±19.80)	84.92 (±6.86)	31.89 (±15.42)	83.77 (±14.50)	91.64 (±6.90)	86.17 (±9.51)
3	69.74 (±6.22)	63.98 (±3.09)	75.30 (±5.31)	68.98 (±14.14)	80.59 (±3.43)	76.58 (±6.44)	56.97 (±9.57)	70.78 (±6.93)	71.87 (±3.92)	15.41 (±5.39)	74.65 (±8.40)	80.26 (±7.34)	**80.78 (±4.68)**
4	**48.16 (±10.15)**	26.8 (±18.88)	22.88 (±10.95)	26.54 (±11.44)	43.38 (±17.09)	38.74 (±10.22)	7.62 (±7.13)	48.14 (±4.33)	13.03 (±9.55)	2.85 (±2.01)	33.07 (±14.92)	19.28 (±5.91)	34.56 (±8.44)
5	78.80 (±4.17)	62.42 (±2.49)	84.42 (±2.17)	86.88 (±2.44)	88.71 (±1.37)	88.83 (±2.26)	79.26 (±6.76)	84.70 (±3.09)	86.10 (±2.66)	50.40 (±2.83)	86.27 (±3.06)	88.95 (±1.64)	**89.26 (±2.14)**
6	94.28 (±2.92)	76.60 (±12.92)	80.80 (±10.67)	91.80 (±5.72)	95.54 (±2.37)	93.89 (±6.67)	87.08 (±6.52)	78.06 (±12.19)	95.09 (±4.63)	47.04 (±6.47)	90.31 (±7.92)	**97.40 (±1.20)**	93.98 (±5.75)
7	89.94 (±0.24)	87.32 (±1.21)	91.00 (±1.09)	91.58 (±1.17)	**94.27 (±1.03)**	93.56 (±1.62)	90.90 (±1.31)	88.07 (±1.68)	91.05 (±1.30)	87.65 (±1.20)	91.75 (±1.96)	93.50 (±0.64)	93.29 (±3.07)
8	87.00 (±1.24)	82.88 (±1.81)	89.04 (±1.54)	88.16 (±3.28)	93.37 (±0.60)	93.67 (±1.72)	89.69 (±0.64)	80.90 (±6.06)	89.57 (±2.25)	82.14 (±0.81)	91.73 (±0.95)	92.44 (±0.63)	**94.37 (±0.83)**
9	74.34 (±3.16)	67.34 (±1.28)	84.30 (±4.60)	80.22 (±4.26)	90.83 (±3.03)	90.22 (±2.85)	81.21 (±1.94)	79.21 (±6.65)	83.15 (±5.03)	49.30 (±3.73)	86.03 (±2.67)	89.31 (±1.86)	**91.07 (±2.54)**
10	85.54 (±6.89)	84.46 (±3.48)	90.10 (±5.84)	89.66 (±3.18)	**93.05 (±1.87)**	90.95 (±5.04)	91.04 (±3.46)	93.17 (±2.80)	92.66 (±2.91)	51.06 (±8.83)	93.04 (±2.05)	89.80 (±4.40)	92.84 (±4.87)
11	96.00 (±3.33)	93.00 (±0.42)	99.36 (±0.72)	98.28 (±1.52)	99.78 (±0.35)	99.84 (±0.30)	99.31 (±0.56)	99.58 (±0.48)	98.73 (±1.06)	98.81 (±0.42)	98.24 (±1.28)	99.49 (±0.68)	**99.95 (±0.10)**
12	80.28 (±20.67)	59.34 (±4.55)	86.58 (±3.82)	92.56 (±7.29)	99.01 (±1.44)	96.87 (±2.11)	89.82 (±8.60)	89.37 (±2.39)	94.61 (±3.74)	88.44 (±4.63)	89.56 (±5.15)	97.62 (±2.89)	**99.46 (±0.60)**
OA	83.89 (±1.52)	78.15 (±0.64)	86.87 (±1.09)	87.37 (±2.13)	92.42 (±0.67)	92.14 (±1.31)	86.40 (±1.74)	84.06 (±3.28)	88.65 (±1.89)	72.49 (±0.96)	89.75 (±1.32)	91.73 (±0.64)	**93.01 (±0.72)**
AA	75.67 (±4.43)	64.13 (±3.21)	84.96 (±2.89)	80.05 (±1.77)	87.08 (±1.06)	86.36 (±1.91)	70.52 (±4.19)	**87.86 (±2.17)**	80.68 (±1.27)	51.23 (±1.94)	82.74 (±1.92)	83.87 (±2.05)	84.99 (±1.97)
k×100	80.04 (±1.85)	76.71 (±8.25)	84.04 (±1.27)	84.41 (±2.51)	90.60 (±0.82)	90.26 (±1.63)	82.99 (±2.23)	80.76 (±3.74)	85.89 (±2.31)	65.54 (±1.04)	87.26 (±1.66)	89.71 (±0.81)	**91.31 (±0.91)**

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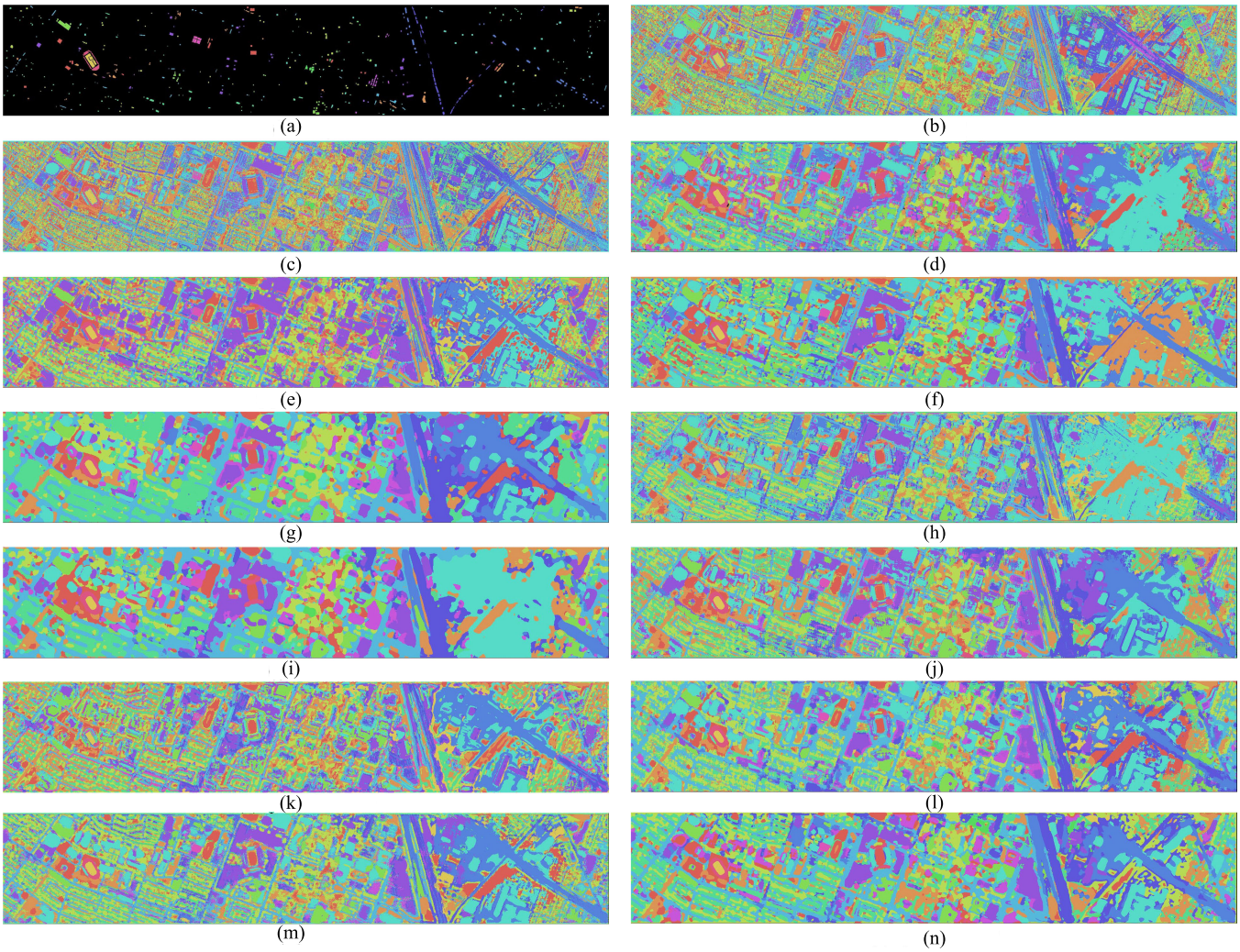


Fig. 16. Classification maps of the Houston 2013 dataset: (a) test ground truth (test gt), (b) SVM (OA = 75.82%), (c) KNN (OA = 74.41%), (d) 2-D-CNN (OA = 87.03%), (e) HybridSN (OA = 81.24%), (f) SSRN (OA = 87.22%), (g) DBDA (OA = 85.83%), (h) RSSAN (OA = 77.82%), (i) FDSSC (OA = 76.18%), (j) ViT (OA = 79.51%), (k) SpectralFormer (OA = 68.89%), (l) RvT (OA = 82.79%), (m) HiT (OA = 84.37%), and (n) HybridFormer (OA=89.52%).

with that described in the original paper. The HybridSN, SSRN, DBDA, RSSAN, FDSSC, SpectralFormer, and HiT, those network structures, were the same as described in the original paper. To provide a fair comparison, all of the methods use the same hyperparameters.

1) Quantitative Comparisons: Tables VIII–XI show the quantitative classification performance of the different algorithms on the four datasets in terms of the OA, AA, and k, in addition to the F1 scores for each category. All results of the ten experiments are listed in the form of mean value $\pm$ standard deviations, and the optimal results of each row are highlighted in bold. Overall, the traditional classifiers, the SVM, and the KNN had the worst classification performance on all datasets, with the exception that they were slightly better than the 2-D-CNN on the Chikusei dataset. Furthermore, the HybridFormer achieved the best accuracy on all of the datasets.

Table VIII lists the classification results on the HS dataset. HybridFormer performs better on most of the land-cover categories, but, for some single categories, the other algorithms are close to or even better than HybridFormer in terms of the F1 score. The principal reason for this result is that the samples in the HS dataset are mostly scattered, small, and localized. However, the HybridFormer still had the highest OA at 89.52%, with a larger improvement between 2.49% and 21.72% than that of the other methods. There are also significant improvements over the other algorithms in terms of the AA and k.

In terms of the HongHu and HyRANK datasets, HybridFormer achieves a definite advantage on all of the evaluation metrics, and it also obtains the highest OAs of 98.83% and 93.01%. As shown in Table IX, the highest accuracy is achieved for almost every class on the HongHu dataset, which further demonstrates the superiority of the HybridFormer algorithm. As shown in Table XI, the CNN-based algorithm achieved better classification accuracy, whereas the SpectralFormer achieved the worst accuracy on the HyRANK dataset due to the fact that, although the transformer structure is able to capture global information, it is still limited in its ability to model local information. In contrast, the proposed HybridFormer model makes full use of the local–global–local strategy to effectively exploit the spatial and spectral

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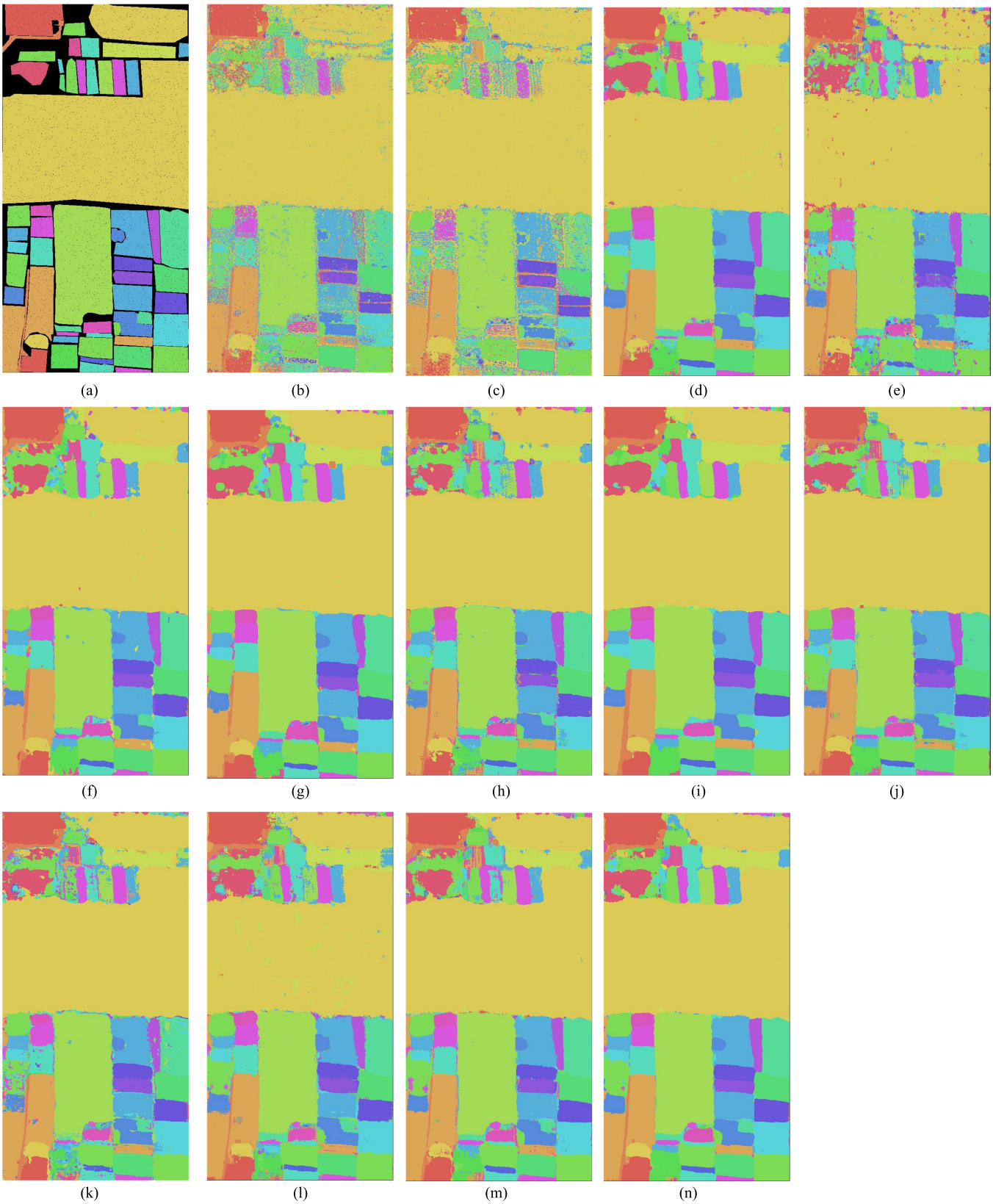


Fig. 17. Classification maps of the WHU-Hi-HongHu dataset: (a) test ground truth (test gt), (b) SVM (OA = 83.17%), (c) KNN (OA = 73.33%), (d) 2-D-CNN (OA = 96.54%), (e) HybridSN (OA = 93.84%), (f) SSRN (OA = 98.39%), (g) DBDA (OA = 98.36%), (h) RSSAN (OA = 94.11%), (i) FDSSC (OA = 98.15%), (j) ViT (OA = 97.08%), (k) SpectralFormer (OA = 94.83%), (l) RvT (OA = 96.95%), (m) HiT (OA = 97.69%), and (n) HybridFormer (OA = 98.83%).

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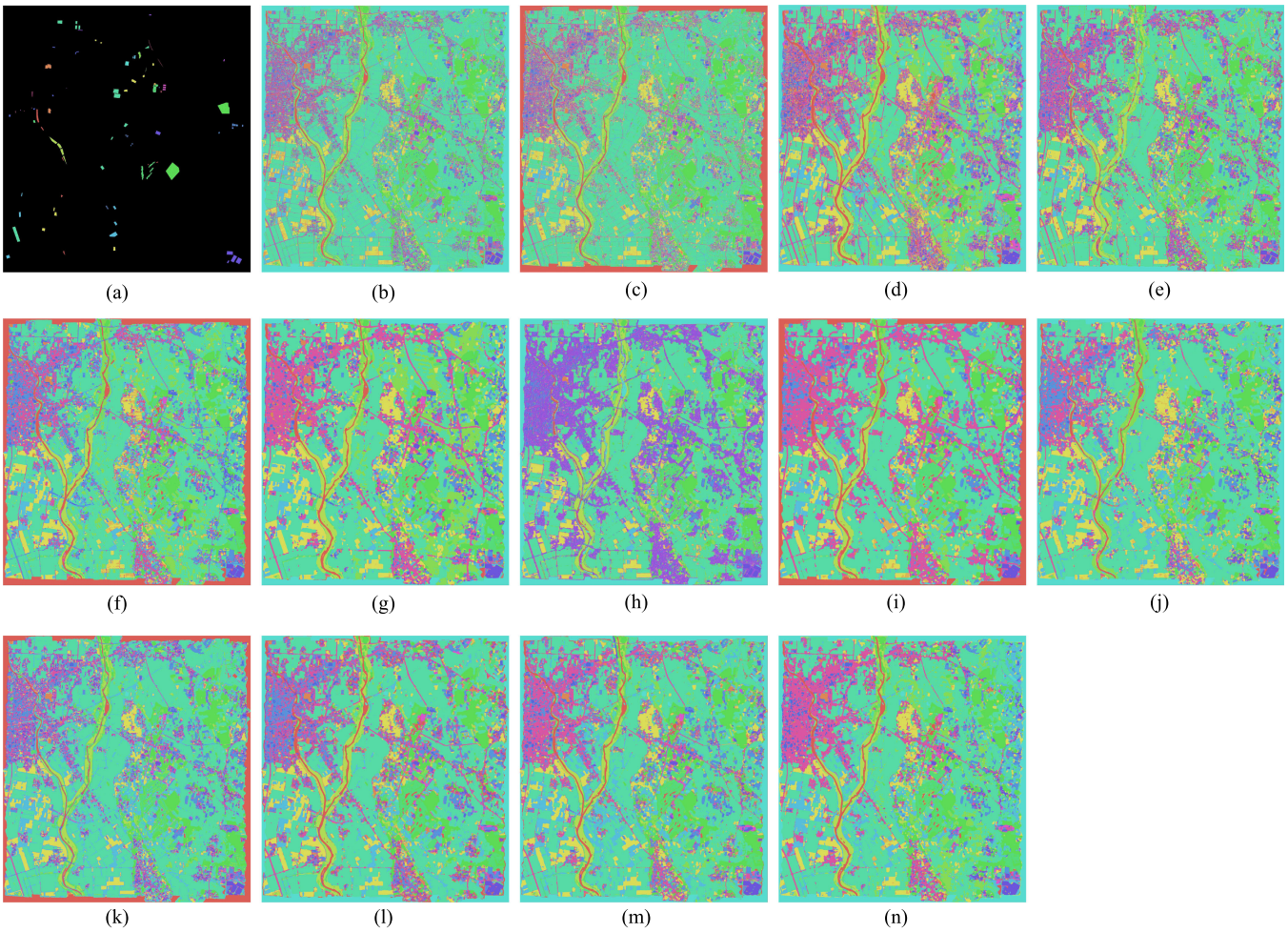


Fig. 18. Classification maps of the Chikusei dataset: (a) test ground truth (test gt), (b) SVM (OA = 97.71%), (c) KNN (OA = 96.99%), (d) 2-D-CNN (OA = 96.32%), (e) HybridSN (OA = 97.29%), (f) SSRN (OA = 99.15%), (g) DBDA (OA = 99.14%), (h) RSSAN (OA = 96.54%), (i) FDSSC (OA = 98.98%), (j) ViT (OA = 98.86%), (k) SpectralFormer (OA = 97.70%), (l) RvT (OA = 96.88%), (m) HiT (OA = 99.18%), and (n) HybridFormer (OA=99.25%).

information of HSI, which dramatically improves the classification performance.

Table X records the classification results for the Chikusei dataset, where the HybridFormer and the HiT performed better than the other algorithms. Specifically, the HybridFormer achieved the highest OA of 99.25%, exceeding the other methods by 0.07%–2.93%. Compared to the other algorithms, the lack of significant accuracy improvements is due to the fact that this dataset has a larger sample size. In summary, the HybridFormer algorithm obtains higher and more even scores on most classes, which also proves the superiority of the algorithm.

2) Qualitative Comparisons: To more intuitively compare the different models, we qualitatively display the classification result plots obtained for the eight algorithms. Figs. 16–19 present the HS, HongHu, Chikusei, and HyRANK datasets, and the different colors correspond to different land covers. By comparing all of the classification result maps with their test ground truth, it is obvious that the classification of the traditional classifier KNN and SVM models produces more noise, and the CNN-based algorithm is prone to misclassification of the boundary pixels between the two land covers. Furthermore, the two transformer-based algorithms, (h) and (i), are prone to classification errors due to the lack of sufficient fusion of local and global information in HSIs. It can be noticed that the proposed HybridFormer method yields more reasonable maps than the other algorithms. Most obviously, it can be seen in Fig. 17 that the HybridFormer can accurately identify almost every category, which more intuitively illustrates its superiority.

3) Complexity and Efficiency Analysis: To further evaluate the effectiveness of the proposed methods, Tables XII and XIII detail the parameter numbers, floating-point operations (FLOPs), training time, and testing time of all methods. As shown in Table XII, HybridFormer has lower FLOPs than most deep learning-based methods and a smaller number of parameters than most transformer-based methods. As shown in XIII, HybridFormer has a moderate training time and testing time. As we can see, the 2-D-CNN has the least training time and test time, the lowest FLOPs, and the number of parameters. This is because the 2-D-CNN has only three 2-D convolutional layers and two fully connected layers. The parameter number of HybridFormer is larger than most of the CNN-based methods, which is due to the fact that its transformer encoder weights of different granularities are not

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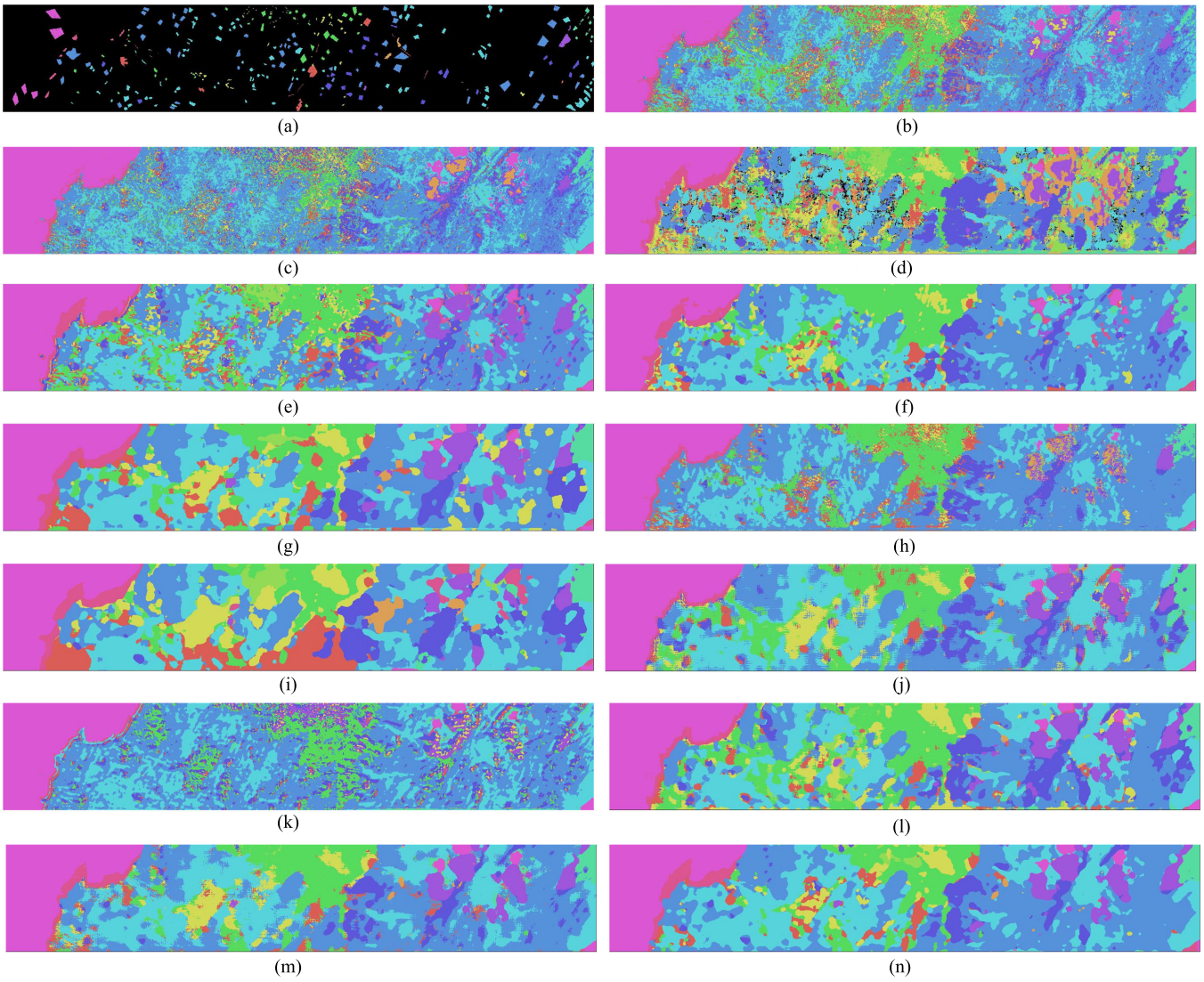


Fig. 19. Classification maps of the HyRANK dataset: (a) test ground truth (test gt), (b) SVM (OA = 83.89%), (c) KNN (OA = 78.15%), (d) 2-D-CNN (OA = 86.87%), (e) HybridSN (OA = 87.37%), (f) SSRN (OA = 90.96%), (g) DBDA (OA = 92.14%), (h) RSSAN (OA = 86.40%), (i) FDSSC (OA = 84.06%), (j) ViT (OA = 88.65%), (k) SpectralFormer (OA = 73.42%), (l) RvT (OA = 89.75%), (m) HiT (OA = 91.73%), and (n) HybridFormer (OA=93.01%).

TABLE XII

FLOPs AND NUMBER OF PARAMETERS OF DIFFERENT METHODS ON FOUR DATASETS

Methods		FLOPs (M)				Number of Parameters (M)			
		HS	HongHu	Chikusei	HyRANK	HS	HongHu	Chikusei	HyRANK
CNN-based methods	2D-CNN	14.35	16.18	14.13	14.81	0.42	0.44	0.42	0.42
	HybridSN	374.59	729.51	329.52	464.73	3.29	5.61	3.00	3.88
	SSRN	315.77	596.43	280.13	387.05	0.28	0.47	0.26	0.33
	DBDA	210.07	398.37	186.15	257.89	0.28	0.51	0.25	0.34
	RSSAN	18.25	26.55	17.23	20.33	0.20	0.30	0.22	0.21
	FDSSC	341.95	651.02	302.7	420.44	0.85	1.61	0.76	1.05
Transformer-based methods	ViT	43.47	49.15	42.76	44.90	6.04	10.44	5.66	6.94
	SpectralFormer	26.95	63.52	23.18	35.05	0.27	0.59	0.24	0.33
	RvT	93.03	96.66	92.57	93.95	3.61	3.75	3.59	3.65
	HiT	1127.26	1572.9	1090.47	1213.46	56.14	64.81	55.43	57.82
	HybridFormer	166.22	512.66	136.44	236.18	1.98	3.58	1.87	2.29

shared, while it still has fewer parameters than HybridSN. Although the HybridFormer is not the optimal method in terms of model efficiency and complexity compared to other methods, it achieves state-of-the-art accuracy among all methods.

4) Visualization: To verify the generalization capability of HybridFormer, different numbers of training samples were randomly selected from the given training set with training sample sizes of 0.8%, 1%, 3%, 5%, and 10%. Fig. 20 represents the ten times the average of OA results obtained by

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TABLE XIII
TRAINING TIME AND TESTING TIME OF DIFFERENT METHODS ON FOUR DATASETS

Methods		Training Time (s)				Testing Time (s)			
		HS	HongHu	Chikusei	HyRANK	HS	HongHu	Chikusei	HyRANK
CNN-based methods	2D-CNN	30.14	1281.00	135.11	45.15	1.32	59.89	6.62	1.92
	HybridSN	47.91	2151.47	211.78	73.91	2.51	129.15	11.85	4.30
	SSRN	68.09	2608.72	260.00	89.34	3.46	179.48	15.89	6.00
	DBDA	65.92	8047.84	303.05	107.57	5.20	254.69	24.24	8.58
	RSSAN	31.31	1325.55	147.08	47.23	1.64	67.86	7.90	2.28
	FDSSC	59.61	19852.74	830.35	103.11	4.11	201.63	19.06	6.75
Transformer-based methods	ViT	31.73	1316.36	144.47	47.28	1.67	67.97	8.17	2.35
	SpectralFormer	75.11	2927.29	272.45	82.79	3.57	521.86	14.46	4.89
	RvT	35.37	1398.49	160.74	51.34	2.21	83.37	10.97	3.06
	HiT	212.84	5516.81	898.78	242.47	43.89	1260.22	222.79	59.71
	HybridFormer	50.24	1763.92	215.96	64.46	3.37	115.12	16.81	4.60

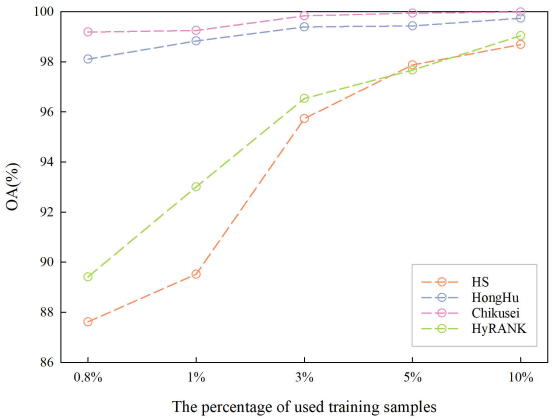


Fig. 20. OA of HybridFormer with different numbers of training samples.

the proposed HybridFormer. The OA results gradually improve the classification performance as the percentage of training samples used increases. In addition, the increase in accuracy tends to flatten as the sample size increases. This reasonable curve further indicates that the HybridFormer model is feasible and stable.

IV. CONCLUSION

In this study, we proposed a hybrid transformer framework with multigranularity SSA for HSI classification. Unlike other transformer-based models that directly flatten image patches, HybridFormer uses convolution to perform shallow feature extraction and multigranularity token embedding. Aiming at the multispectral dimensional nature of HSIs, the SSA-based multibranch transformer encoder can capture the global relationship between different tokens, which enables the model to focus on more discriminative features in the spatial domain and the spectral domain. Furthermore, an adaptive fusion approach is designed to balance the complementary information found on different branches. The experimental results on the four datasets demonstrate the superiority of the HybridFormer model. In the future, we will further explore how transformer-based structures can make full use of information from the HSI with smaller samples, including unsupervised learning and self-supervised learning. The architecture will be further optimized for the high-dimensional HSI features so that a lightweight framework can be designed.

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